

Tianzhe Chu

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EDUCATION

- **The University of Hong Kong** Hong Kong SAR
Ph.D. student at IDS advised by Prof. Yi Ma
Sep 2024 - Now
Research: Foundation Models, Deep Representation Learning, Reinforcement Learning, Agent.
- **ShanghaiTech University** Shanghai, China
B.Eng in Computer Science and Technology (Honor)
Sep 2020 - Jun 2024
Core Courses: Machine Learning, Probability and Statistics, Computer Architecture, Data Structure and Algorithms, Discrete Mathematics, Signal Processing, Mathematical Analysis
- **University of California, Berkeley** Berkeley, CA, US
Visiting undergraduate in EECS
Aug 2022 - May 2023
Core Courses: Deep Learning, Deep Reinforcement Learning, Foundation of Graphics, Applications of Parallel Computing, Computer Vision, Optimization Model in Engineering

RESEARCH INTEREST

- Foundation models, world models: data, architecture, training, and understanding.
- Application of multimodal models in agents and robotics.

PUBLICATIONS

(* means equal contribution)

- **Tianzhe Chu***, Yuexiang Zhai*, Jihan Yang, Shengbang Tong, Saining Xie, Dale Schuurmans, Quoc V.Le, Sergey Levine, Yi Ma, *SFT Memorizes, RL generalizes: A Comprehensive Study of Foundation Model Post-training*, <https://arxiv.org/abs/2501.17161v1>,
ICML 2025
- Yaodong Yu*, **Tianzhe Chu***, Shengbang Tong, Ziyang Wu, Druv Pai, Sam Buchanan, Yi Ma, *Emergence of Segmentation with Minimalistic White-Box Transformers*, <https://arxiv.org/abs/2308.16271>.
Conference on Parsimony and Learning (CPAL) 2024 (Oral), <https://cpal.cc>,
NeurIPS 2023 XAI Workshop (Oral), (4 orals of 59 posters)
- **Tianzhe Chu***, Shengbang Tong*, Tianjiao Ding*, Xili Dai, Benjamin D. Haeffele, René Vidal, Yi Ma, *Image Clustering via the Principle of Rate Reduction in the Age of Pretrained Models*, <https://arxiv.org/abs/2306.05272>,
ICLR 2024
- Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Benjamin D. Haeffele, Yi Ma, *White-Box Transformers via Sparse Rate Reduction*, <https://arxiv.org/abs/2306.01129>.
NeurIPS 2023
- Yaodong Yu, Sam Buchanan, Druv Pai, **Tianzhe Chu**, Ziyang Wu, Shengbang Tong, Hao Bai, Yuexiang Zhai, Benjamin D. Haeffele, Yi Ma, *White-Box Transformers via Sparse Rate Reduction: Compression Is All There Is?*, <https://arxiv.org/abs/2311.13110>.
Journal of Machine Learning Research (JMLR)
- Chun-Hsiao Yeh*, Chenyu Wang*, Shengbang Tong, Ta-Ying Cheng, Ruoyu Wang, **Tianzhe Chu**, Yuexiang Zhai, Yubei Chen, Shenghua Gao, Yi Ma, *Seeing from Another Perspective: Evaluating Multi-View Understanding in MLLMs*, <https://arxiv.org/abs/2504.15280>,
Tech Report

RESEARCH EXPERIENCE

- **HKU Musketeers Foundation Institute of Data Science** Hong Kong SAR
Graduate research assistant advised by Prof. Yi Ma
Sep 2024 - Now
- **Berkeley Artificial Intelligence Research (BAIR) in UC Berkeley** Berkeley, CA, US
Undergraduate research assistant advised by Prof. Yi Ma
Nov 2022 - Jun 2024

SELECTED AWARDS

- **Outstanding Graduate** Shanghai, China
Affiliation: ShanghaiTech University Jul, 2024
- **Merit Student** Shanghai, China
Affiliation: ShanghaiTech University Dec 2023
- **Provincial First Prize for 35th National Physics Olympics Competition** Nanjing, Jiangsu, China
Affiliation: Suzhou High School of Jiangsu Province Sep 2018

INVITED TALKS

- SFT Memorizes, RL Generalizes @ Cathy Wu’s lab, MIT, Apr. 2025
- SFT Memorizes, RL Generalizes @ BAAI, Apr. 2025
- SFT Memorizes, RL Generalizes @ CPAL 2025, Stanford, Mar. 2025
- Presented on White-Box Transformers @ SVIP Lab, ShanghaiTech, Sep, 2023
- Presented on Emergence of Segmentation in White-Box Transformers @ CPAL 2024, HKU, January, 2024